Busy Making Other Plans:

A Simulation Study on the Effects of Deviations from Preregistrations

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The inability to reliably reproduce findings in the field of psychology is a known problem. The Open Science Collaboration (2015) found that only 36% of significant results were reproducible. One reason for this "replication crisis" is "researcher degrees of freedom" (Simmons et al., 2011). This term describes the flexibility that researchers have in decisions made during their work. This includes decisions such as how many observations to collect, which statistical model to use and which covariates to add. When these decisions are made during data collection or analysis, researchers can make opportunistic choices which steer their results towards desired outcomes. Such opportunistic choices are called *p*-hacking and can dramatically increase false-positive rates (Stefan & Schönbrodt, 2023).

The preregistration was proposed as a way to limit these types of practices. Preregistrations are documents that are published at the start of the research process, in which decisions concerning data collection and analysis are recorded. Templates are available to researchers, to inform them about what information is important to preregister. The preregistration is then uploaded to a public repository, with a time stamp of when it was published. Peer reviewers, and others, have access to the preregistration and can evaluate it before the start of the study or compare it to the final research paper. This practice increases transparency about research choices and has become more popular, with the number of preregistrations increasing annually (Ferguson et al., 2023; Lindsay et al., 2018).

In theory preregistration works well, if researchers are specific in their preregistration and follow it closely, this should lead to better reproducible findings. In practice, preregistrations are often imprecise and incomplete, making them less effective at restricting researcher degrees of freedom. Furthermore, no difference has been found between preregistered and non-preregistered papers in the number of significant findings (van den Akker, van Assen, et al., 2024), and no reduction in *p*-hacking (Brodeur et al., 2024).

Another common problem are deviations from preregistration (Claesen et al., 2021; van den Akker, Bakker, et al., 2024). Deviating means that the preregistered plan is not being followed during the execution of the study. Deviations have been found in as many as 93% of preregistered papers, with up to 89% being incompletely reported (Claesen et al., 2021; Willroth & Atherton, 2024). These changes occur most commonly in data collection procedures, statistical models and exclusion criteria (van den Akker, Bakker, et al., 2024). This includes differences in the total sample size, how outliers are selected and which covariates are used. These deviations could serve to further diminish the effectiveness of preregistrations, as it reintroduces researcher degrees of freedom and leaves room to revert back to *p*-hacking strategies.

Are all deviations bad?

Deviations from preregistrations occur in all different aspects of the research process, but what is not known is the effect of these deviations. Do changes from original research plans reduce research quality? Lakens (2024) theorized what the possible consequences could be. If deviations occur due to assumptions being violated or data no longer being suitable for the planned analysis, then deviating from the original research plan can have positive effects on the validity of a test. Similarly, adding additional analyses can increase the robustness. On the other hand, changes in testing also mean that a hypothesis is generally less “severely” tested. This means that a theory has not been given the proper opportunity to be falsified which can result in higher type II errors. Consequently, little can be said about whether deviations are bad, especially when the reason for the deviation is unknown.

There is no clear consensus on whether inconsistencies between preregistrations and final publications are a problem. When asked, researchers themselves deemed deviations to be problematic to varying degrees (Willroth & Atherton, 2024). Changes to analyses and hypotheses were considered the least acceptable, whereas changing research platforms or software were deemed relatively justifiable. The estimated impact on the results and whether the deviation was reported transparently and were also taken into consideration. To summarize, even though deviations from preregistration are common, a clear conclusion on their effects is still missing.

The present project

With this simulation study, I intend to close this gap in the literature and explore the effects of typical preregistration deviations on research outcomes and the quality of published results. Specifically, I aim to answer the question: ‘How do deviations from preregistrations affect type I and type II error rates?’ The effect of deviations is studied in three domains: sample size, outlier exclusion criteria, and the statistical model. Different deviations are simulated in each domain and compared to a baseline condition without deviations. This research question is investigated in the context of social sciences and uses a linear regression as the model of interest. The effects are examined across two scenarios: a no-effect scenario, and an effect scenario, in which the main effect parameter is changed. The aim of this study is descriptive and therefore no specific hypotheses are formulated.

As a secondary research question, I address the issue of reasons for deviations being unknown. The reason behind a deviation is important for assessing its effect. A researcher being forced to deviate due to circumstance is very different from a researcher *choosing* to deviate. When a researcher chooses to deviate, this could be because they are picking the results they prefer. Initially, the deviation conditions are simulated and the outcomes are used to assess type I and type II error as compared to the baseline condition. For the second research question, an ‘opportunistic’ condition is created in which outcomes are selectively chosen from either the baseline or the deviation conditions, based on which condition has a significant p-value. This is done in order to proxy the opportunistic selection of results by the researcher.

With these simulations I hope to shed light on the effect of deviations from preregistration on type I and type II errors, as well as the effect of picking deviations opportunistically.

Methods

Conditions

I reviewed which deviations to investigate by (1) how common they are, (2) their potential impact and (3) their justifiability. Based on this, deviations were chosen in the following domains: sample size, outlier exclusion criteria, and statistical model. Within the three chosen domains, multiple conditions were simulated and compared to a baseline condition without any deviations.

Sample size

Deviations in the sample size are some of the most common deviations, with consistency between preregistrations and published papers only being 28% for the exact sample size (van den Akker, Bakker, et al., 2024). Changing the sample size can inflate type II error (by decreasing power) as well as the type I error (Lakens, 2024; Simmons et al., 2011). Small deviations in sample size are common and often due to factors outside of the researcher's agency. Researchers also admit to stopping collection earlier or continuing collection longer after finding disappointing results (John et al., 2012). However, it is unknown how often this is also used as a reason to deviate from preregistrations. Based on this, the conditions in Table 1 are simulated within the sample size domain.

Outlier exclusion criteria

Literature shows that over half of published papers do not adhere to their pre-specified outlier exclusion criteria (Claesen et al., 2021; van den Akker, Bakker, et al., 2024). Vague or missing criteria in preregistrations make it harder to assess deviations and leave a lot of room for interpretation and ad hoc decisions (Heirene et al., 2024). This could potentially lead researchers to choose how to exclude outliers based on which method provides better results. Researchers themselves, however, deem outlier deviations to be relatively acceptable (Willroth & Atherton, 2024).

Lakens (2024) argues that adding additional exclusion criteria can undermine the severity of a test. This idea is corroborated by Stefan & Schönbrodt (2023), who showed that the type I error increases linearly with the number of outlier detection methods used. Furthermore, 38% of researchers admit to excluding outliers only after examining their impact on the data (John et al., 2012). If this is also the reason why many people deviate from their preregistered outlier criteria, then these deviations are important.

The most common method for identifying outliers in the social sciences is the z-score (Bakker & Wicherts, 2014; Leys et al., 2013). A minimum of three standard deviations from the mean is often maintained as the rule of thumb for excluding cases. Within linear regression, outliers are also often identified based on their influence, commonly using Cook's distance. Cook's distance excludes outliers based on how much their exclusion would influence the regression coefficients. In this study, the baseline condition will be to exclude datapoints based on a z-score of 3 standard deviations from the mean, with deviation conditions based on a stricter z-score and Cook's distance (Table 1).

Statistical model

Amongst all of the aforementioned domains, the least is known about why people deviate from their preregistered statistical model. "Statistical model" is a very broad domain and includes many types of deviations, such as changes in the dependent variable, independent variable, statistical inference criteria and the actual statistical test. In this paper, "statistical model" refers to the above-mentioned types of changes, not only the statistical test and its specifications.

The broadness of the domain also means that deviations within the domain have been investigated from multiple angles. Claesen et al. (2021) found a deviation rate of 70% within the "analysis" domain, including deviations like examining additional effects, changing which model is tested and performing unregistered robustness checks. The majority of these deviations went unreported. Another study reported inconsistencies in 40% of the statistical models, deviations included the specifications of the variables, which model was tested and how variables were used (van den Akker, Bakker, et al., 2024).

As Lakens (2024) argues, at times it might be necessary to alter the statistical model. This could happen when a variable has been measured at a different level than expected or a test assumption has been violated. However, changing the statistical model can also be done opportunistically. Choosing to add an extra dependent variable because results are not yet satisfying can increase the type I error rate to 9.5%, and the addition of a covariate can increase it to 11.7% (Simmons et al., 2011). The reasoning behind the change thus becomes very important. Therefore, deviations due to failing to properly specify how variables would be operationalized in the preregistration are also seen as a larger shortcoming than model deviations due to unforeseen consequences (Willroth & Atherton, 2024). Within the statistical model, three deviation conditions are simulated: outcome switching, adding a continuous covariate, and adding a dichotomous covariate (Table 1).

Table 1: Parameter Values for Baseline Condition and Deviation Conditions per Domain

<table>
<tr><th>Domain</th><th>Baseline condition</th><th>Deviations conditions</th></tr>
<tr><td>Sample size</td><td>200</td><td>+5
+30
-5
-30</td></tr>
<tr><td>Outlier exclusion criteria</td><td colspan="2">Z-scores of 3 SD | Z-scores of 2 SD |
Cook's distance of 1 |</td></tr>
<tr><td colspan="3">Statistical methods: covariates | No covariates | Continuous covariate | | | | | Dichotomous covariate</td></tr>
<tr><td colspan="3">Statistical methods: outcome switching | Y | Ya |</td></tr>
</table>

Note. Variable Ya is defined in the next section.

Data-generating mechanism

Data is generated under two scenarios: "X has an effect on Y" and "X has no effect on Y". The only difference in data generation between the two scenarios is in the main effect parameter, β_1. In the effect scenario, β_1 is 0.2, and in the no-effect scenario, β_1 is 0. In the baseline condition, the statistical model is defined as

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

where Y is the dependent variable, β_0 the intercept, β_1 the main effect, and ϵ the random error.

The population parameter values are presented in Table 2, together with the parameter values for the deviation conditions. The complete data-generating model is defined as

$$\begin{aligned} Y &= \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 D + \epsilon, \\ Y_a &\sim Y, \quad \rho(Y, Y_a) = 0.6, \end{aligned}$$

where Z and D represent a continuous and categorical covariate respectively and Y_a represents an alternative outcome variable with a correlation of 0.6 with variable Y.

As mentioned, data is generated under two scenarios: an effect scenario and a no-effect scenario. The only difference in data generation between the two scenarios is in the main effect parameter, β_1.

In order to assess differences between outlier exclusion criteria, the data needs to include outliers. These are simulated by generating 5% of the data, 2.5% at each end of the distribution, with -3 and 3 as mean values for the normal distribution from which ϵ is drawn.

Table 2: Parameter Values for the Data generating mechanism

Parameter	Value
Intercept	$\beta_0 = 0$
Regression coefficient	$\beta_1 = 0$ or $\beta_1 = 0.20$
Independent variable	$X \sim \mathcal{N}(\mu, \sigma^2)$
Random error	$95\% = \epsilon \sim \mathcal{N}(0, 1)$
	$2.5\% = \epsilon \sim \mathcal{N}(-3, 1)$ |
	$2.5\% = \epsilon \sim \mathcal{N}(3, 1)$ |
Continuous covariate regression coefficient	$\beta_2 = 0.06$
Continuous demographic variable	$Z \sim \mathcal{N}(\mu, \sigma^2)$
Dichotomous covariate regression coefficient	$\beta_3 = 0.06$
Dichotomous demographic variable	$D \sim \text{Bernoulli}(0.5)$
Alternative outcome error	$\epsilon_a \sim \mathcal{N}(\mu, \sigma^2)$
Sample size	200

Parameter	Value

Estimands

The estimand of this study is β_1, which represents the effect of the independent variable X on dependent variable Y. The coefficient is estimated using the `stats` package in `R` (R Core Team, 2024), under the baseline and deviation conditions.

Outcomes

As previously mentioned, there are two scenarios under which data is generated: a no-effect scenario and an effect scenario. Within these scenarios, each simulated data set is examined under different conditions. The conditions include a baseline condition, with no deviations, and one condition for each possible deviation (Table 1).

To answer the main research question, the type I and type II error rates are compared between the baseline result and each deviation result. All ten conditions, one baseline and nine deviation conditions, are executed using the same data set. In each repetition the β_1 regression coefficient, the p-value and the confidence interval are recorded for each condition.

To answer the second research question, the effect of deviating opportunistically is examined. This is done using the same data set as in the first section. For each repetition, the baseline p-value is assessed. When it is significant, the baseline condition is recorded. If it is non-significant, the deviation conditions of that repetition are checked. If any of the deviation conditions are significant, one is chosen at random and recorded for that repetition. As a result, for each repetition, a significant result will be recorded if it is available among any of the conditions. This provides an 'optimized' data set. The error rates will then be compared between the baseline results and the opportunistic result. An overview can be found in Figure 1.

Note. This figure illustrates the study phases. In phase 1, the baseline result is compared to the deviation result. In phase 2, the baseline result is compared to the best result.

Performance measures

Performance is assessed through the type I and type II error for the estimand β_1. Specifically, the regression coefficient itself, the p-value and the confidence interval are recorded for each condition in each repetition.

The type I error refers to a false-positive, or rejecting the null-hypothesis when it is true. In this study that would mean detecting an effect in the no-effect scenario. Type I error is generally acceptable at a rate of 5% or below, based on an alpha level of .05. In this case, a rate of higher than 5% is considered an inflated type I error rate.

Type II error refers to a false negative, or failing to reject the null-hypothesis when it is false. In this study that would mean failing to detect an effect in the effect scenario. Type

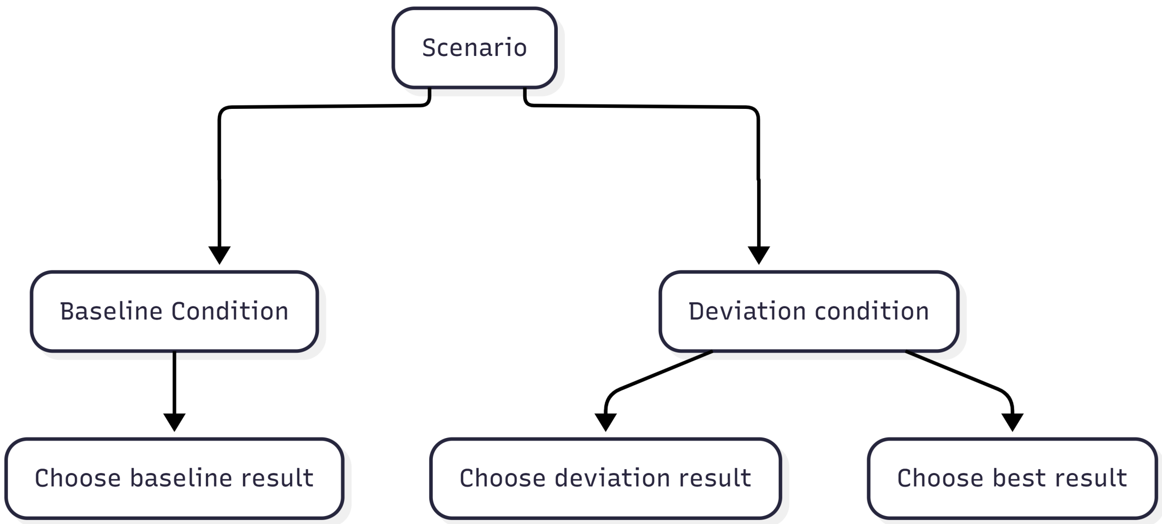

Figure 1: Condition Comparisons

II error is commonly assessed in the form of power. Power is $1-\beta$, where β is the type II error. The nominal type II error rate of .20 results in a generally accepted power of 80%. In this study, type II error will be expressed in the form of power rates and consequently, power below 80% is considered deflated power.

The number of iterations is based on the desired precision of the project. A Monte-Carlo standard error (MCSE) of 0.01 is deemed appropriate. According to the formula by Morris et al. (2019), the required number of iterations to achieve the desired MCSE is 1600 (Equation 1). With an MCSE of 0.01, power rates are assessed at a precision interval of [79, 81] around the nominal power of 80%. Power levels outside of this range can be stated to have increased or deflated power levels that are not due to simulation noise.

For the type I error, 1600 iterations results in an MCSE of 0.00545 (Equation 2). This means that type I error is assessed with a precision interval of [0.045, 0.055] around the nominal α level of .05. Any type I error rates outside of this interval are considered deflated or inflated.

Disclosures

Preregistration, reproducibility and ethics

This project was preregistered on the 31th of March, 2026. The preregistration can be found at [link]. In order to prevent "seed hacking", I preregistered a seed which would only become available after the date of preregistration. An R package was created for this project under the name `thepack` in which all functions necessary to simulate the data can be found. The package is publicly available at [githublink]. In order to advance reproducibility, all other files related to this project are publicly available at [github link]. This includes data, code

$$
\begin{aligned}
MCSE &= \sqrt{\frac{\widehat{\text{Power}}\left(1-\widehat{\text{Power}}\right)}{n_{\text{sim}}}},\\
0.01 &= \sqrt{\frac{\widehat{0.8}\left(1-\widehat{0.8}\right)}{n_{\text{sim}}}},\\
n_{\text{sim}} &= \frac{0.8\times 0.2}{(0.01)^2},\\
n_{\text{sim}} &= 1600
\end{aligned} \tag{1}
$$

$$
\begin{aligned}
MCSE &= \sqrt{\frac{\widehat{\text{Type I error}}\left(1-\widehat{\text{Type I error}}\right)}{n_{\text{sim}}}},\\
MCSE &= \sqrt{\frac{\widehat{0.05}\left(1-\widehat{0.05}\right)}{1600}},\\
MCSE &= \sqrt{2.96875e-05},\\
MCSE &= 0.00545
\end{aligned} \tag{2}
$$

Figure 2: *Note.* Calculations for the number of iterations based on the desired MCSE for power and the MCSE for type I error based on the number of iterations.

scripts and output files. I made use of `renv` to create a reproducible environment in which package versions and R version are recorded. Instructions on how to reproduce the findings can be found in the README file on GitHub.

Ethics approval was obtained on October 7th, 2025 from the Utrecht University faculty of Social Sciences under case-number #25-1980.

Results

Research question one

The primary research aim of this study is to examine the effects of deviations from preregistration on type I and type II errors. In this section the type I error and power of each deviation condition are compared to the nominal levels. The Monte Carlo standard error resulted in a precision range of [0.045, 0.055] around the nominal type I error of 0.05 and a range of [79,81] around the nominal 80% power. Values outside of these ranges are considered inflated or deflated. Figure 3 and Figure 4 show the type I error and power for each condition along with the precision range.

Effects on power

Within the Sample Size domain, the effects reflect what is known from the existing literature. Increasing the sample size, increases the power and decreasing the sample size decreases the power [sources]. These effects can be observed in the current study as well (Figure 3).

Deviations in the Outlier Exclusion Criteria domain also impact power. When using a strict *z*-score (exclusion based on 2 *SD* (standard deviation) instead of 3 *SD*) power increases to 83.8%. This means that we are able to detect existing effects more often than in the nominal rate. When using Cook's distance instead of *z*-scores, the power drops to 68.2%. This indicates that, when this method is used, we are less likely to observe an existing effect in the generated data then we would in the nominal rate. Cook's distance is generally considered a less conservative form of outlier exclusion because it only excludes observations based on their influence on the regression coefficient. It is therefore quite unexpected to see such a large drop in power in this condition.

In the Statistical Model domain, deflated power is only observed in the alternative outcome condition. Deviating by using an alternative, correlated outcome variable resulted in a power of 35.2%. Both the continuous and dichotomous covariate conditions have no apparent effect on power.

This simulation study demonstrates that power can be affected when researchers are forced to deviate from a preregistration. This effect can be positive or negative, depending on the type of deviation. Overall, collecting much fewer participants, using Cook's distance as outlier criterium and switching to an alternative outcome all deflate powers. Power is most impacted by switching outcomes, with a steep drop off from the nominal 80% to 35.2% power. Small changes in sample size and the addition of a covariate show no effect, whereas collecting many more participants than planned and using a *z*-score of two standard deviations show slight increases in power.

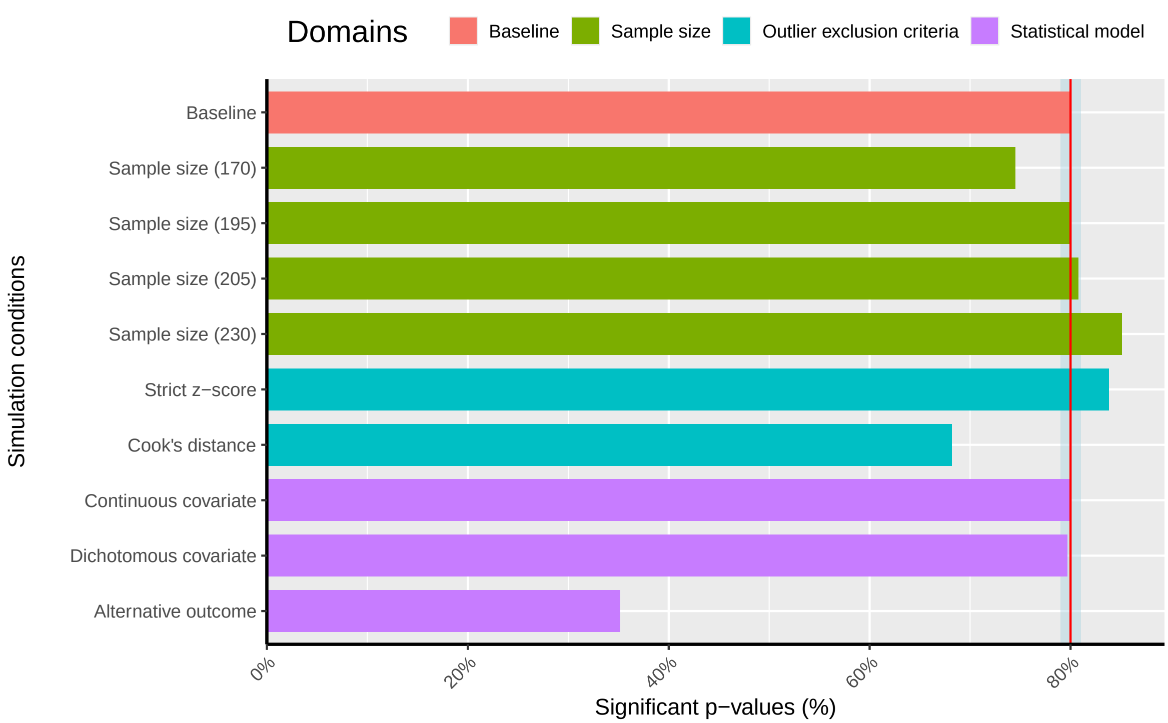

Figure 3: Power for Baseline and Deviation Conditions in the Effect scenario

Effects on Type I error

Only two conditions show results outside of the precision range. Small increases in effect size show an increased type I error of 5.6%. Using a strict *z*-score results in a type I error rate of 5.8%. All other deviation scenarios fall within the prespecified precision range and therefore do not indicate any effects on the type I error (Figure 4).

The small inflation when slightly increasing the sample size is a borderline case. Using the MCSE of 0.055, which was preregistered for this study, means that it falls outside of the precision range. Outside of this range the effects are unlikely to be due to simulation noise. Using the empirical MCSE of this condition, however, shows that the found effect might still be due to noise. With a type I error of 0.055625, the empirical MCSE becomes 0.00573. This means that type I error rates higher than 0.055729 would be considered inflated, which this condition just remains beneath. In light of this, all other conditions were also evaluated using the empirical MCSE. The slight increase in sample size remained the only condition in which this changed the conclusion.

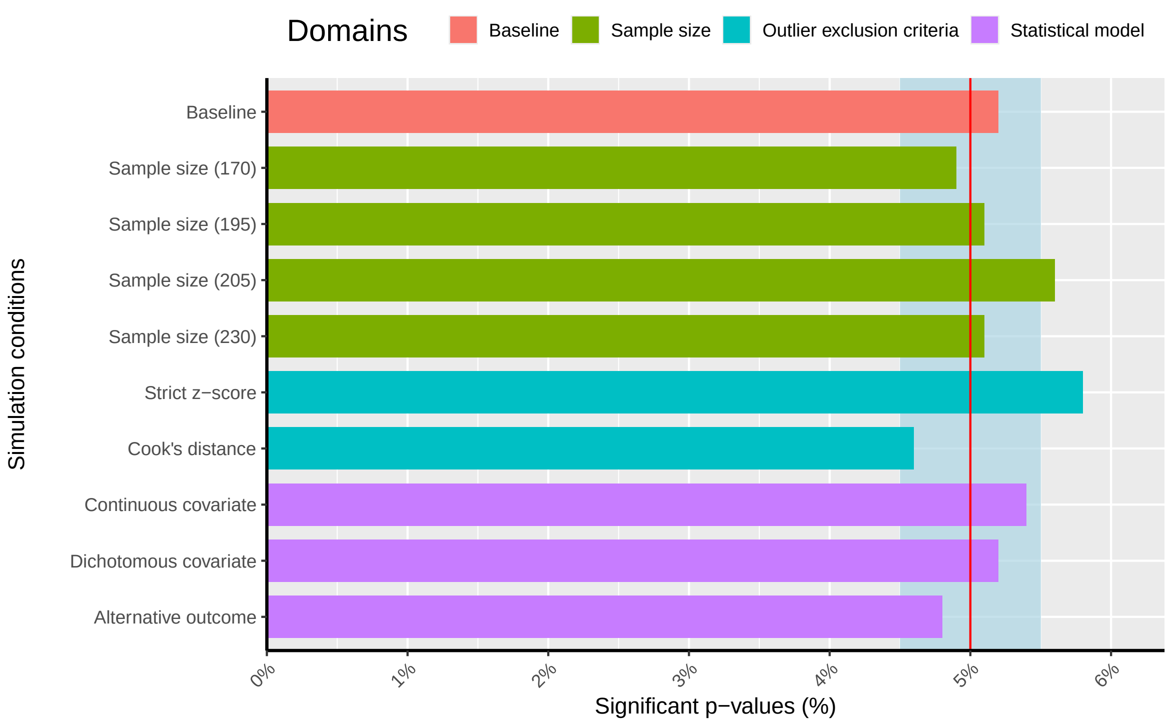

Figure 4: Type I error rate for Baseline and Deviation scenarios in the no-effect Scenario

Research question two

The second research question pertains to the effect of *choosing* to deviate when the baseline condition is non-significant. This is done by comparing an opportunistic scenario to the nominal type I error and power. In the opportunistic scenario, one condition was recorded per iteration. Whenever the baseline condition was significant, it was selected. If the baseline was non-significant, a significant deviation condition was selected instead, if available, and the number of significant deviations was recorded. Table 3 displays how often significant baselines and significant deviation conditions were found.

As expected, we observed a positive effect of choosing to deviate on power. Conditions were selected in order to maximize the number of significant results, consequently, an increase in power was anticipated. In 1484 out of 1600 repetitions, a significant condition was observed. This results in a power of 92.7% in the effect scenario. In the no-effect scenario, 261 out of 1600 repetitions showed a significant condition. Yielding an inflated type I error rate of 16.3%. This is a large inflation compared to the nominal level of 5% and similarly much higher than the highest type I error rate observed in a singular deviation condition. The highest type I error rate observed for research question one was 5.8% in the Strict z-score condition.

In the effect scenario, 79.9% of the repetitions resulted in a significant baseline. Of the 321 repetitions where the baseline was non-significant, 36.1% also reported no significant deviation conditions. The remaining 64% of repetitions, in which the baseline was non-significant, had at least one significant deviation, with seven significant deviation conditions at most.

In the no-effect scenario, the majority (83.7%) of the repetitions showed a non-significant baseline as well as non-significant deviations (Table 3).

Significant baseline conditions were observed 5.2% of the time, which is in line with the nominal α level of 5%. However of the repetitions in which the baseline was non-significant, 13.3% showed at least one significant result. This indicates that when researchers obtain non-significant results from their preregistered data, and choose to deviate in one of the investigated ways, they can still find a significant result 13.3% of the time. When combining the significant baseline conditions and significant deviation conditions, the type I error rate inflated to 16.3%. This is a large increase in the number of false positives when picking an available significant result.

Discussion

Claesen et al. (2021) and van den Akker, Bakker, et al. (2024) have shown that deviating from preregistration plans happen very commonly. The goal of this simulation study was to investigate the effects of these deviations. In order to do so we identified three main domains in which deviations commonly occur and put together nine deviation conditions. These deviations were then examined in terms of their effects on type I and type II error. The results of the study show that both are affected by deviations.

Forced deviations

The results of this simulation study show that type I error and power are both affected when researchers are forced to deviate from a preregistration. Power can be affected both positively and negatively, depending on the type of deviation and whether there is an effect to be

Table 3: Distribution of the number of significant deviations (with a non-significant baseline

Significant deviation conditions	Occurences	
	Effect	No-effect
(Significant baseline)	1279 (79.9%)	83 (5.2%)
0	116 (7.2%)	1339 (83.7%)
1	75 (4.7%)	126 (7.9%)
2	61 (3.8%)	27 (1.7%)
3	30 (1.9%)	17 (1.1%)
4	31 (1.9%)	5 (0.3%)
5	4 (0.2%)	1 (0.1%)
6	3 (0.2%)	2 (0.1%)
7	1 (0.1%)	0 (0%)

detected. This corroborates the idea theorized by Lakens (2024) that it is not inherently bad to deviate from a preregistration. Power showed small inflations and deflations, depending on the type of deviation. Only two deviation conditions showed effects on the type I error. For all other conditions, effects on type I error were not strong enough to suggest an effect outside of simulation noise. This shows that there is little risk of inflating the type I error when being forced to deviate from a preregistration except for using a stricter z-score as outlier criterium.

Before conducting research, the researcher does not know whether they will be working with an effect or not. As a result, the researcher cannot predict whether their deviation will hurt their type I or type II error. Deviating in the number of standard deviations used to calculate z-scores can positively impact power in an effect scenario yet inflate type I error in a no-effect scenario. Because the researcher does not know which scenario they are facing, this type of deviation is best avoided in order to prevent an inflated type I error rate in the case of a no-effect scenario.

Consequently, from these results we can conclude that negative impacts on type I and/or type II can occur when researchers are forced to deviate by 1) strongly decreasing the sample size, 2) using Cook's distance as outlier criterium (as opposed to a z-score of 3 *SD*), 3) switching to an alternative outcome with a moderate to high correlation with the original outcome variable, and 4) using a stricter z-score as outlier criterium. These four types of deviations appear to be most consequential as they can either deflate power or inflate the type I error.

Opportunistic deviations

The results also showed an effect of deviating opportunistically on the type I error rate and on power. The type I error rate of 16.3% in the opportunistic condition indicates that when researchers choose to make alterations to their methodology when they find non-significant results, their risk of reporting false positives becomes much higher.

The types of deviations investigated in this simulation are based on which deviations are commonly reported. Many other deviations might also occur, and many deviations might also go unreported. This means that even in preregistered papers, type I error rates might be even higher than the 16.3% found in this study and much higher than the aspired 5%.

Limitations

It is important to note that the data simulated in this study was based on one set of parameter values. This is likely to make the results less generalizable to other disciplines in which the currently used data structure is less common. The baseline sample size, effect size and methods investigated are all common among social sciences but might look much different for other statistics oriented disciplines such as medicine. It remains unknown whether the found effects would hold up under smaller and larger samples or effect sizes. Therefore, future research might explore whether these results persist under alternative data-generating mechanisms.

Furthermore, opportunistic deviation was limited to singular deviations in this research, meaning that only one deviation was made at a time, deviations were not be combined. In vivo, researchers might choose to deviate in multiple ways simultaneously. Similar to *p*-hacking, multiple deviation methods might be combined to achieve significant results. This could lead to even higher type I error rates than found in this study. These results might therefore underestimate the actual impact of opportunistic deviations and real type I error rates in deviation papers might be a lot higher. Further research into this topic might explore a factorial version of this simulation study to examine these effects.

Implications and conclusions

Deviating from a preregistered research plan is not inherently bad. It can serve to improve or worsen the number of false positives and negatives depending on the type of deviation and the reason for deviating. Forced deviations show negative impacts on power and type I error but might be hard to prevent due to the uncontrollable nature of the deviation. Choosing to deviate in order to get better results shows a larger impact on the number of false positives and therefore forms a bigger issue.

As of yet, researchers often fail to report their reasons for deviating (Claesen et al., 2021; Willroth & Atherton, 2024). This is a big issue when the current results show that choosing to deviate opportunistically leads to a much larger inflation in the type I error rate than forced deviations. There has been a call to report deviations from preregistrations for a while now (Claesen et al., 2021; Heirene et al., 2024; Willroth & Atherton, 2024). We suggest going one step further, and recommend reporting the reasons for the deviation as well. A deviation template has been proposed by Willroth & Atherton (2024) which includes reporting the reasons for deviating. We recommend that this type of reporting becomes the standard among preregistered papers. When researchers properly document how they deviate and why they deviated, it would allow readers to evaluate the possible effects of the deviations more critically. Readers can then draw their own conclusions on the validity of the results.

The goal of preregistration is to limit flexibility and improve reliability of research. The lack of transparency in the reasons for deviations limits the extent to which these goals can be achieved. Therefore we recommend that researchers register the reasons for deviating from their preregistrations. Because deviations are not inherently bad, but they do have the potential to be harmful.

Transparency *Author Contributions* Anna J. Vijlbrief: Conceptualization; Methdology; Investigation; Formal Analysis; Software; Visualization; Writing - original draft; Writing - reviewing and editing Anne M. Scheel: Conceptualization; Supervision; Writing - reviewing and editing Hanne I. Oberman: Conceptualization; Supervision; Writing - reviewing and editing

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